

HumanMaximizer AI Lead Generation

Architecture & Design Document

Razor Infotech Pvt Ltd — AI Architect / GenAI Engineer Assignment

Prompt Examples

HumanMaximizer AI Lead Generation System

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All prompts are Jinja2 templates in backend/prompts/. They use StrictUndefined mode — any missing variable raises an error at render time rather than silently producing a hallucination-prone prompt.

1. ResearchAgent — research_summary.j2

Purpose

Generates a 3-paragraph structured company intelligence report that the QualificationAgent and SalesAgent consume downstream.

Template Structure

You are an expert B2B sales researcher for an HRMS software company.

COMPANY INFORMATION:
Company: {{ company_name }}
Domain: {{ domain }}
Industry: {{ industry }}
Employees: {{ employee_count or "Unknown" }}
Location: {{ hq_location or "Unknown" }}
Current HR Tech: {{ tech_stack | join(", ") if tech_stack else "Unknown" }}
Growth Signal: {{ growth_signal | replace("-", " ") | title }}

WEBSITE CONTENT (scraped):
{{ description[:2000] if description else "No website content available." }}

Write a 3-paragraph lead summary using ONLY the facts above. Do not invent statistics, metrics, or content.

Format your response exactly as follows:

COMPANY OVERVIEW: [One paragraph summarising the company's business, scale, and market position]

HR PAIN POINTS: [One paragraph describing specific HR, payroll, or workforce challenges this company likely faces]

HRMS FIT SIGNAL: [One paragraph explaining why or why not this company is a good candidate for HumanMaximizer]

Example Input

Company: Metropolis Healthcare Ltd
Domain: metropolishealthcare.com
Industry: Healthcare
Employees: 4,000
Location: Mumbai, Maharashtra
Current HR Tech: Excel, Google Sheets
Growth Signal: Expansion (entering tier-2 cities)
Website Content: "Metropolis Healthcare operates 220+ diagnostic labs across India... recently expanded to Patna, Bhopal, and Nagpur... managing 4,000+ employees across clinical and administrative roles..."

Example Output (Mistral-7B)

COMPANY OVERVIEW: Metropolis Healthcare Ltd is one of India's leading diagnostic chains, operating 220+ labs across major cities and now expanding aggressively into tier-2 markets. With 4,000 employees spanning clinical staff, phlebotomists, lab technicians, and administrative teams, Metropolis operates a complex multi-location workforce that spans multiple states.

HR PAIN POINTS: Managing HR for 4,000 employees across 220+ labs using Excel and Google Sheets creates significant operational risk. Every statutory filing cycle – PF, ESIC, PT across multiple states – requires manual reconciliation. As Metropolis expands into tier-2 cities, each new location adds a new state's compliance requirements, multi-site attendance tracking complexity, and onboarding volume that spreadsheet-based systems cannot handle efficiently.

HRMS FIT SIGNAL: Strong HRMS candidate. Metropolis's spreadsheet-based HR is a critical signal – at 4,000 employees across 220+ multi-state labs, they are well past the threshold where manual HR creates measurable compliance risk and cost. The tier-2 expansion directly increases this complexity. HumanMaximizer's multi-state statutory compliance module and multi-location attendance management directly address Metropolis's stated pain points.

2. QualificationAgent — qualification_reasoning.j2

Purpose

Generates a 2-sentence qualification reasoning that explains the strongest signal and the biggest risk — used by sales reps to understand the opportunity at a glance.

Template Structure

You are a B2B sales qualification analyst for HumanMaximizer HRMS.

LEAD DATA:

Company: {{ company_name }}
Industry: {{ industry }}
Employees: {{ employee_count or "Unknown" }}
Location: {{ hq_location or "Unknown" }}
Tech Stack: {{ tech_stack | join(", ") if tech_stack else "Unknown" }}
Growth Signal: {{ growth_signal | replace("_", " ") | title }}
Decision Makers: {{ decision_makers | length }} found

SCORE BREAKDOWN:

- Company Size Fit: {{ score_breakdown.company_size_fit }}/20
- Industry Relevance: {{ score_breakdown.industry_relevance }}/20
- Tech Stack Gap: {{ score_breakdown.tech_stack_gap }}/20

```
- Decision Maker Reachability: {{ score_breakdown.decision_maker_reachability }}/20
- Growth Signal: {{ score_breakdown.growth_signal }}/20
Total Score: {{ qualification_score }}/100
```

Write exactly 2 sentences:

1. The strongest qualification signal (why this lead is a good HRMS prospect)
2. The biggest risk or blocker for this opportunity

Be specific. Reference the actual score values and company data.

Example Input

Company: Metropolis Healthcare Ltd
Industry: Healthcare
Employees: 4,000
Tech Stack: Excel, Google Sheets
Growth Signal: Expansion

Score Breakdown:

```
- Company Size Fit: 20/20
- Industry Relevance: 20/20
- Tech Stack Gap: 18/20
- Decision Maker Reachability: 0/20
- Growth Signal: 20/20
Total: 78/100
```

Example Output (Mistral-7B)

Metropolis Healthcare's Excel-based HR for 4,000 employees across 220+ labs is the strongest HRMS fit signal in this dataset (18/20 tech gap), compounded by active tier-2 city expansion adding new state compliance obligations at exactly the point where manual systems break down. The primary risk is zero decision-maker reachability (0/20) – no CHRO, HR Director, or People Manager contact was identified, making multi-channel outreach impossible without a warm introduction or LinkedIn search.

3. SalesAgent — cold_email.j2

Purpose

Generates a personalized B2B cold email grounded exclusively in retrieved HumanMaximizer product knowledge. The template explicitly forbids hallucination.

Template Structure

You are a B2B sales copywriter for HumanMaximizer HRMS.

COMPANY PROFILE:

```
Company: {{ company_name }}
Industry: {{ industry }}
Employees: {{ employee_count or "Unknown" }}
Location: {{ hq_location or "Unknown" }}
Current HR Tech: {{ tech_stack | join(", ") if tech_stack else "Unknown" }}
Growth Signal: {{ growth_signal | replace("-", " ") | title }}
{% if decision_makers %}
Decision Maker: {{ decision_makers[0].name }}, {{ decision_makers[0].title }}
{% endif %}
```

QUALIFICATION SUMMARY:

```
{{ qualification_reasoning }}
```

HUMANMAXIMIZER PRODUCT KNOWLEDGE (RAG – use ONLY these facts):

```
{% for chunk in rag_chunks %}
```

```
---
```

```
{{ chunk }}
```

```
{% endfor %}
```

```
---
```

Write a personalized cold email using ONLY the facts above. Do not invent statistics, percentages, customer names, or product features not mentioned in the knowledge above.

Requirements:

- Subject line: specific to this company, ≤ 8 words
- Body: 150-220 words
- Tone: professional but conversational, peer-to-peer
- Reference specific HR pain points from the company profile
- Mention 1-2 specific HumanMaximizer features from the RAG knowledge above
- End with a single, low-friction call to action (15-minute demo)

Format exactly as:

SUBJECT: [subject line]

BODY:

[email body]

Example Output (Mistral-7B)

SUBJECT: Automate Multi-State HR Compliance at Metropolis Healthcare

BODY:

Dear [CHRO Name],

Managing HR for 4,000 employees across 220+ labs in Excel means every PF, ESIC, and PT filing cycle across multiple states is a manual reconciliation exercise – and your tier-2 expansion into Patna and Nagpur just added three new compliance jurisdictions.

HumanMaximizer's statutory compliance module handles PF, ESIC, and PT filings across all states automatically – generating forms, calculating deductions, and filing on schedule from a single dashboard, without reconciliation.

Our multi-location attendance management integrates across all your lab locations into one view, so payroll runs on clean data from day one of every expansion.

Healthcare chains similar to Metropolis – with distributed multi-state workforces – are exactly the use case we've built for.

Would you have 15 minutes this week for a demo tailored to multi-location diagnostics operations? I can walk through the compliance automation end to end.

Best regards,

HumanMaximizer Sales Team

4. SalesAgent — linkedin_message.j2

Purpose

Generates a short, high-conversion LinkedIn connection request message. Strictly limited to ≤ 300 characters to fit LinkedIn's DM character limit.

Template Structure

```
You are a B2B sales outreach specialist.

RECIPIENT: {{ dm_name }}, {{ dm_title }} at {{ company_name }}

PAIN HOOK (the key HR challenge for this company):
{{ pain_hook }}

HUMANMAXIMIZER ONE-LINER (use this exact product description):
{{ hm_one_liner }}

---

Write a LinkedIn connection request message. Requirements:
- Maximum 300 characters total
- No emojis
- No exclamation marks
- Start with "Hi {{ dm_name_first }}, "
- Reference the specific pain hook
- Mention HumanMaximizer in one sentence
- End with an open question
```

Example Input

```
Recipient: Priya Mehta, CHRO at Metropolis Healthcare
Pain Hook: Managing HR for 220 labs through Excel is creating compliance risk as you expand to tier-2 cities
HM one-liner: HumanMaximizer automates multi-state statutory compliance for Indian enterprises with dist
```

Example Output (Mistral-7B)

```
Hi Priya, scaling Metropolis across 220 labs while managing multi-state compliance on
Excel must be creating real pressure as you enter tier-2 markets. HumanMaximizer automates
this end-to-end. Would you be open to a quick chat?
```

(≤ 300 characters: 264)

5. SalesAgent — self_critique.j2

Purpose

A second Mistral-7B pass that verifies the generated email contains no claims unsupported by the provided RAG chunks. Emits structured JSON for the Prometheus hallucination counter.

Template Structure

```
You are a fact-checker for B2B sales emails.

GENERATED EMAIL:
{{ email_body }}

APPROVED PRODUCT FACTS (from HumanMaximizer knowledge base):
{% for chunk in rag_chunks %}

---
```

```
{{ chunk }}
{% endfor %}
```

Review the email above. Identify any specific claims, statistics, percentages, customer names, or product features that are NOT supported by the approved facts above.

Respond in JSON format only:

```
{
  "hallucinated_claims": ["<claim 1>", "<claim 2>"],
  "is_grounded": true or false,
  "verdict": "<one sentence summary>"
}
```

If all claims are supported, return:

```
{
  "hallucinated_claims": [],
  "is_grounded": true,
  "verdict": "All claims verified against product knowledge base."
}
```

Example Output (Mistral-7B)

```
{
  "hallucinated_claims": [],
  "is_grounded": true,
  "verdict": "All claims verified against product knowledge base – statutory compliance module, multi-l"
}
```

Summary

Template	Agent	Model Calls	Temperature	Context Budget
research_summary.j2	ResearchAgent	1	0.2	~2,000 tokens input, ~400 output
qualification_reasoning.j2	QualificationAgent	1	0.1	~800 tokens input, ~80 output
cold_email.j2	SalesAgent	1	0.3	~3,000 tokens input, ~250 output
linkedin_message.j2	SalesAgent	1	0.3	~600 tokens input, ~60 output
self_critique.j2	SalesAgent	1	0.0	~2,500 tokens input, ~100 output

Total LLM calls per qualified lead: 5 (4 generation + 1 critique)

Total LLM calls per disqualified lead: 2 (research summary + qualification reasoning)